9.7 COMPILERS AND INTERPRETERS

We studied, till date the computer knows only one language,i.e., machine language. But we know only High Level Languages and Application Languages, which the computer does not know. But, we both want to interact. How this become possible?

Example:

- *Assume that our prime minister is going to RUSSIA to meet the Russians president.*
- *Our PM knows Hindi and English. And the Russian president knows only RUSSIAN language.*
- *But, both want to converse.*
- *How?*
- *You can see a man (translator) who knows both Russian and English will sit near to them. He will translate English to Russian and vice versa.*

Just as like the above, we need a translator to translate our High Level Languages and Application Languages into corresponding machine language.The compilers and interpreters are here used as translator.

Oh….fine…

- Whether it is software or hardware?

Absolutely, it is a software written in one particular high level language.

Then, why we should need both, compiler and interpreters?

To understand this let we see the working method of complier and interpreter.

Compiler: Compiler means taking together. This will take entire program (whatever be the number of lines) and convert it into machine language. Wherever the mistake, just it will mark it and go to next line. After completing the entire program, it will list out the errors at the last.

Interpreter: The interpreter will read line by line. If there is a mistake in the first line, it will stop there and the error will be displayed. It can go to the next line, only after the particular line gets corrected.

This is the difference between Compiler and Interpreter.

NOTE: Along with each High Level Language and Application language either Compiler or Interpreter will be attached. Widely compilers are used.

9.8 OPERATING SYSTEM

The word system denotes computer. Hence Operating System means operating computer. In this, we are going to see actually how the computer gets operated.

Before going further, recollect the following.

- Basically computer never do anything on its own.
- ROM, RAM (Internal Memory).
- Hard disc (External Memory)… etc.

Operating System helps us to do some important and vital operations (like copy, delete, etc.) before entering into any particular language and after coming-out. It forms a bridge in between the user and the computer.

ELEC_EQUIP_Chapter09.indd 196 11/10/2020 11:53:06 AM